H

millennia institute

Class:	Adm No

Candidate Name: _____

2026 Preliminary Examinations
Pre-University 3

GEOGRAPHY (HIGHER 2)
Paper 2
INSERT

9173/02
14 September 2026

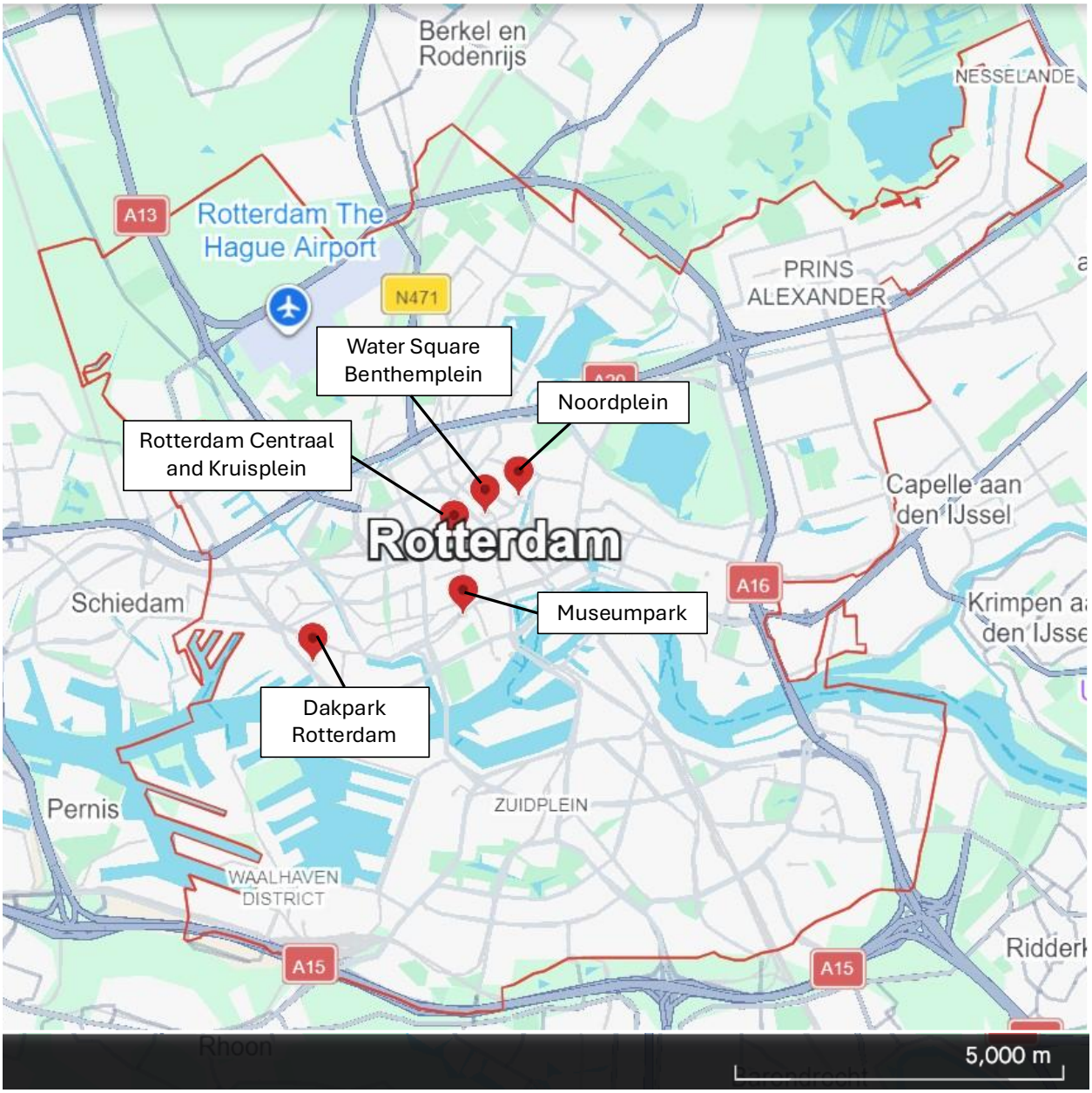
3 hours

READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

Resource 1 for Question 1

Locations of the five fieldwork sites in Rotterdam, Netherlands



*The red line denotes the perimeters of the fieldwork site.

Resource 2 for Question 1

Basic information about the public measures implemented at the five locations

Location	Basic information	Climate-adaptation measures
Water Square Benthemplein	A public square used for sports and recreation. It is surrounded by schools, businesses and housing.	Three sunken basins collect and temporarily store about 1.7 million litres of rainwater, reducing pressure on drains.
Rotterdam Centraal and Kruisplein	A busy transport area used by commuters, residents and tourists. It has extensive paved surfaces.	An underground system stores excess water from the nearby canal during heavy rainfall.
Noordplein	A public square surrounded by housing, shops and roads. It is used for markets and community events.	Trees and planted areas intercept some rainfall, but much of the square is paved and depends on conventional drains.
Dakpark Rotterdam	An elevated public park built above a shopping complex. It contains lawns, gardens and recreational spaces.	Vegetation and soil retain some rainwater, slow surface runoff and provide cooling and shade.
Museumpark	A public park and cultural area containing museums and an underground car park.	An underground reservoir can temporarily store about 10 million litres of rainwater, reducing flood risk.

Resource 3 for Question 1

Photographs of Water Square Benthemplein during wet (right) and dry (left) weather

The Water Square Benthemplein in Rotterdam uses three concrete basins to hold nearly 1.7 to 2 million liters of rainwater during storms. In dry weather, these sunken pits function as public sports and social spaces. During wet weather, they transform into visible retention lakes that protect the city from floods.



Dry Weather Function

- **Sports and Play:** The largest deep basin acts as a basketball and soccer court.
- **Skating and Wheels:** One shallower basin function as a skate area with steel gutters.
- **Gathering Places:** The third basin provides an event stage or amphitheater seating for people to relax.



Wet Weather Function

- **Water Collection:** Stainless steel channels guide rainwater runoff from the square and nearby roofs straight into the basins.
- **Gradual Filling:** Two shallow basins fill during regular rain, while the deep sports pit fills only during heavy or continuous downpours.
- **Slow Release:** Water stays stored safely above ground until the city system can drain it away without flooding.

Resource 4A for Question 1

Survey questionnaire used by the students

Community responses to climate change in Rotterdam

Location: _____ Date: _____ Time: _____

1 What is your connection to Rotterdam?

- ☐ Live here
- ☐ Work here
- ☐ Study here
- ☐ Visiting

2 What is your age group?

- ☐ 18–25
- ☐ 26–40
- ☐ 41–60
- ☐ Over 60

3 What is your approximate monthly household income?

- ☐ Below €2,000
- ☐ €2,000–€3,999
- ☐ €4,000–€5,999
- ☐ €6,000 or above

4 Before today, had you heard of pluvial flooding and blue-green infrastructure?

- ☐ Yes, both
- ☐ Only pluvial flooding
- ☐ Only blue-green infrastructure
- ☐ No, neither

5 How concerned are you about the effects of intense rainfall in Rotterdam?

Not concerned 1 2 3 4 5 **Extremely concerned**

6 To what extent do you support the following public climate-adaptation measures?

Measure	Strongly oppose	Oppose	Neither	Support	Strongly support
Water squares	1	2	3	4	5
More parks and green spaces	1	2	3	4	5
Permeable pavements	1	2	3	4	5
Underground water storage	1	2	3	4	5

7 Should Rotterdam spend more public money on these measures because they reduce flooding and improve residents' lives?

- ☐ Yes
- ☐ No

8 Which household climate-adaptation measures have you adopted? Tick all that apply.

- ☐ Installed a green roof
- ☐ Installed a rain barrel
- ☐ Replaced impermeable paving
- ☐ Planted more vegetation
- ☐ Moved valuable possessions above possible flood level
- ☐ Purchased flood insurance

9 Approximately how much have you spent on household climate-adaptation measures?

- ☐ Nothing
- ☐ Below €500
- ☐ €500–€1,999
- ☐ €2,000 or above

10 What else should the community do to prepare for intense rainfall?

Resource 4B for Question 1

Environmental perception survey on climate-adaptation measures

Location: _____ **Date:** _____ **Time:** _____ **Weather:** _____

Observe the climate-adaptation measures within approximately 50 metres. Circle one score for each characteristic.

Negative perception	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Positive perception
Measures are difficult to see	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Measures are highly visible
Purpose of measures is unclear	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Purpose of measures is clear
Measures appear ineffective	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Measures appear effective
Measures are poorly maintained	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Measures are well maintained
Site is unattractive	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Site is attractive
Measures are inaccessible	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Measures are accessible
Site is rarely used	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Site is frequently used
Measures do not improve the site	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Measures greatly improve the site
Site appears vulnerable to rainfall	−5	−4	−3	−2	−1	0	+1	+2	+3	+4	+5	Site appears resilient to rainfall

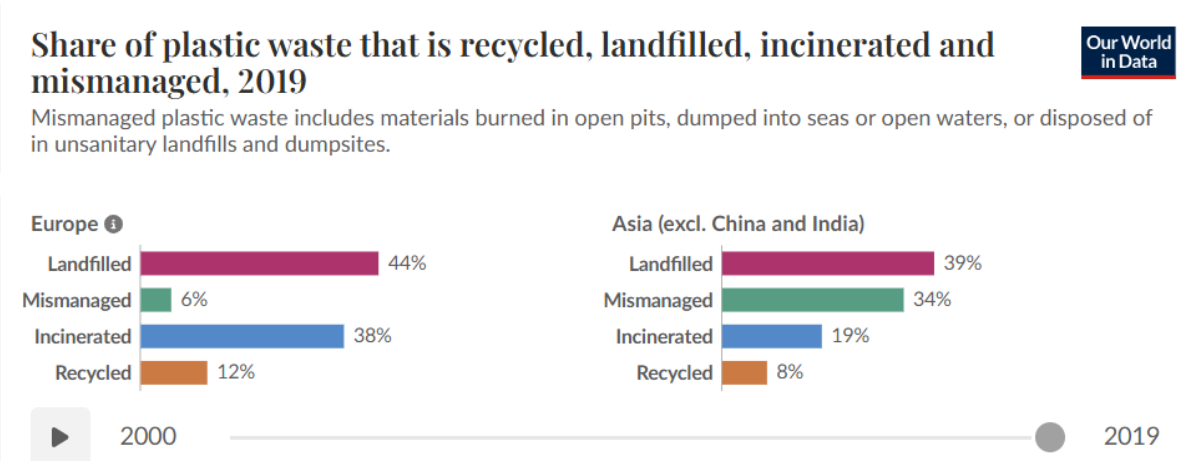
Resource 5 for Question 1

Data collected by the students for environmental perception survey on climate-adaptation measures

Element assessed	Benthemplein Water Square	Rotterdam Centraal and Kruisplein	Noordplein	Dakpark Rotterdam	Museumpark
Visibility of adaptation measures	+4.3	−2.8	−3.5	+2.8	−3.8
Clarity of purpose	+4.0	−3.0	−3.7	+1.2	−3.0
Perceived effectiveness	+4.2	+0.5	−1.8	+2.0	+1.5
Maintenance	+3.2	+3.5	+2.0	+3.8	+4.0
Attractiveness	+3.5	+2.8	+2.5	+4.5	+4.3
Accessibility	+4.0	+4.5	+4.0	+4.2	+4.0
Community use	+3.7	+4.8	+4.2	+4.0	+4.2
Improvement to the site	+4.0	+1.0	+0.8	+4.3	+3.0
Resilience to intense rainfall	+4.3	+1.5	−1.0	+2.5	+2.0
Total mean score	**+35.2**	**+12.8**	**+3.5**	**+29.3**	**+16.2**

Resource 6 for Question 2

Share of plastic waste management methods in 2019 in Europe and Asia in 2019



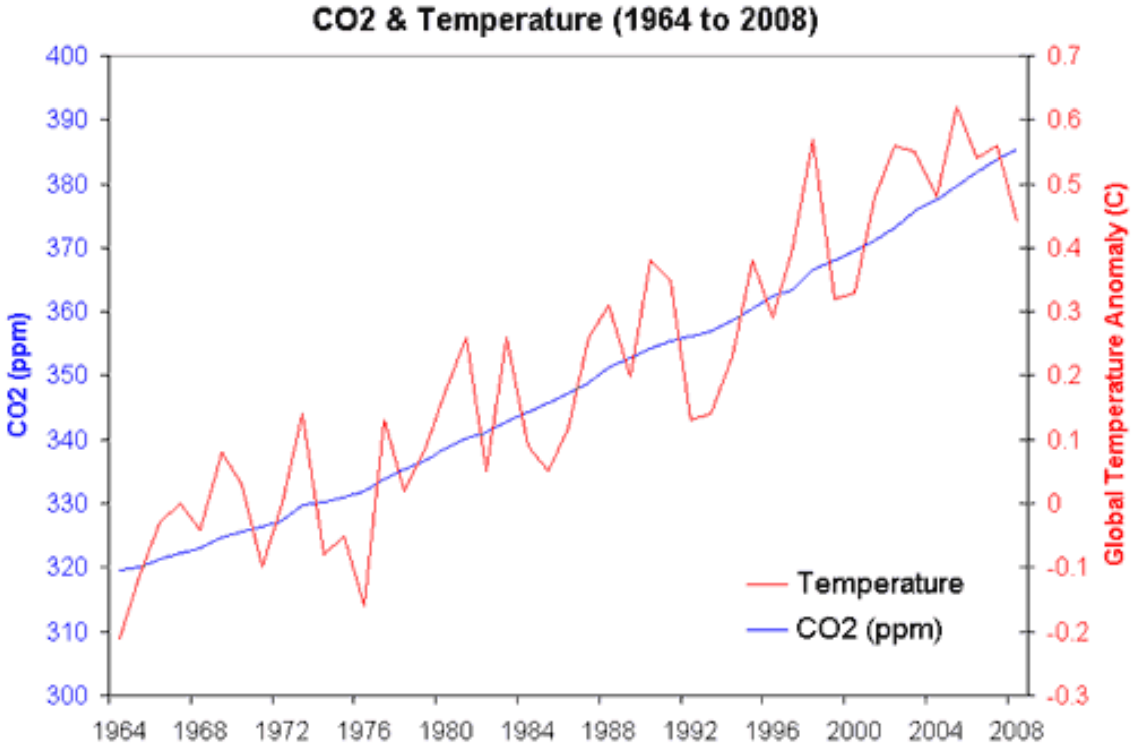
Resource 7 for Question 2

Smokey Mountain slum in Manila, Philippines



Resource 8 for Question 2

Atmospheric CO_2 (parts per million, NOAA) and Global Temperature Anomaly (°C, GISS) from 1964 to 2008.



Resource 9 for Question 2

Flooding in the Bengal Delta, Bangladesh, in April 1991.



Copyright Acknowledgements

Resource 1	Google maps (author's own)
Resource 2	https://landezine.com/water-square-benthemplein-by-de-urbanisten/ (Last accessed 26 July 2026)
Resource 3	https://www.urbanisten.nl/work/benthemplein https://www.iurc.eu/wp-content/uploads/2022/05/Rotterdam-Presentation_compressed.pdf https://www.rotterdam.info/en/visit/finder-locations/rotterdamse-oogstmarkt https://www.rotterdam.info/en/visit/finder-locations/dakpark https://paulderuiter.nl/en/projects/parkeergarage-museumpark (Last accessed 1 August 2026)
Resource 4A and 4B	Author's own
Resource 5	Author's own
Resource 6	https://ourworldindata.org/grapher/share-plastic-fate?time=2019 (Last accessed 19 August 2026)
Resource 7	https://www.justonewayticket.com/smokey-mountain-a-walk-through-the-slums-of-manila-philippines/ (Last accessed 19 August 2026)
Resource 8	https://skepticalscience.com/the-correlation-between-co2-and-temperature.html (Last accessed 19 August 2026) https://www.lse.ac.uk/granthaminstitute/publication/tackling-flooding-in-bangladesh-in-a-changing-climate/ (Last accessed 19 August 2026)
Resource 9	https://ourworldindata.org/grapher/share-plastic-fate?time=2019 (Last accessed 19 August 2026)